

Feed Forward Neural Network

1. Aim

To implement a **Feed Forward Neural Network (FFNN)** for classifying input data by passing information through interconnected layers of neurons from the input layer to the output layer without forming any feedback connections.

2. Problem Statement

A Feed Forward Neural Network is an Artificial Neural Network used to learn patterns from input data and make predictions or classifications. The problem is to train a neural network using given input and output data so that it can learn the relationship between them and correctly predict the class of new input data.

3. Solution

The problem is solved by constructing a neural network consisting of an **input layer, one or more hidden layers, and an output layer**. The input values are passed forward through the network, where weights and activation functions are used to calculate the output. The network learns by adjusting its weights during training to reduce the prediction error. After training, the model is used to classify new input data.

4. Algorithm

1. Initialize the input dataset and corresponding target values.
2. Create the input, hidden, and output layers of the neural network.
3. Initialize the weights and biases.
4. Pass the input values through the network using **forward propagation**.
5. Apply an activation function to the neurons.
6. Calculate the predicted output.
7. Compare the predicted output with the actual output.
8. Calculate the error using a suitable loss function.
9. Update the weights and biases using **backpropagation** and an optimization method.
10. Repeat the training process for several epochs until the error is reduced.
11. Provide new input data to the trained network.
12. Display the predicted class or output.

5. Program

```
from sklearn.neural_network import MLPClassifier
```

```
# Training data
```

```
X = [
```

```
[0, 0],  
[0, 1],  
[1, 0],  
[1, 1]  
]
```

```
# Target values
```

```
y = [0, 1, 1, 0]
```

```
# Create Feed Forward Neural Network
```

```
model = MLPClassifier(  
    hidden_layer_sizes=(4,),  
    activation='relu',  
    solver='adam',  
    max_iter=2000,  
    random_state=42  
)
```

```
# Train the model
```

```
model.fit(X, y)
```

```
# Test data
```

```
test_data = [[1, 1], [0, 1]]
```

```
# Prediction
```

```
prediction = model.predict(test_data)
```

```
print("Test Data:", test_data)
```

```
print("Predicted Output:", prediction)
```

6. Result

The Feed Forward Neural Network was successfully implemented using Python and Scikit-learn. The network learned the relationship between the input features and target values during training and successfully generated predictions for the given test data.

Output:

Test Data: [[1, 1], [0, 1]]

Predicted Output: [0 1]

Thus, the **Feed Forward Neural Network successfully learned from the training data and predicted the appropriate output for the given test inputs.**

